Model Selection

Econ 524

November 12, 2020

Introduction

For many problems, we may estimate a variety of models to answer our question.
We need some way to compare models and to evaluate which model is best.
Outline

- Attempt 1: goodness of fit measures like training MSE (mean squared error), Log Likelihood, R^2
- Attempt 2: adjustments to goodness of fit measures (AIC, BIC, adjusted R^2, likelihood ratio tests)
- Training MSE vs Test MSE
- Attempt 3: Cross Validation

Attempt 1: training MSE, Log Lik

- MSE is an intuitive way to compare the goodness of fit of various models to learn which model best explains the data.
- $MSE = \frac{1}{n}\Sigma_i^n(y_i - \hat{f}(x_i))^2$
- If we calculate MSE using the training data (the same data we estimated the model with), then I will call it the "training MSE."
- Adding a parameter to a model cannot make the training MSE increase. Ex. in linear model, could set parameter to 0 to get MSE without the parameter.
- MSE has a problem if used on its own to compare models because of "overfitting" – i.e. that adding parameters to a model will make it fit better on the specific data ("training data") but may not improve performance out-of-sample.

Attempt 1: training MSE, Log Lik

- Like MSE, the log likelihood (LL) of a model evaluated at the estimated parameter values provides some info on how well the model fits the data.
- The higher the log likelihood, the more likely it is that we observed the data if the model is the true model, and thus the better the model explains the data.
- Also like MSE, adding parameters to model can never make LL decrease and so it is not an effective way to compare models.

Attempt 2: adjustments to MSE, LL

- A simple fix to the problem described above is to use training MSE or LL but adjust them to account for number of parameters.

AIC

Akaike Information Criterion (AIC)

- AIC is an alternative to LL (or training MSE) that can be used to compare models.
- It is based on the max LL (evaluated at parameter estimates) but includes a penalty for models with more parameters to deal with the overfitting problem.
- $AIC = -2LL + 2k$ where k is the number of parameters
- Models with smaller AIC are better.
- Note that $2k$ term is the "penalty" for increasing the number of parameters.

AIC

Akaike Information Criterion (AIC)

- In the case of linear regression with normal error, the formula can be expressed in terms of training MSE:
- $AIC = nlog(MSE) + 2k$ where n is number of observations
- Note: this formula will provide different values than the other one because they are off by a constant term. But that constant term does not matter for comparison of models.

BIC

Bayesian Information Criterion (BIC)

- BIC is an alternative to AIC that has a greater penalty for models with more parameters.
- $BIC = -2LL + klog(n)$ where k is the number of parameters and n is number of observations
- Models with smaller BIC are better.
- Since $log(n)$ is greater than 2 for $n \geq 8$, the "penalty" for increasing the number of parameters is larger with BIC than AIC.
- Like AIC, we can rewrite the formula in terms of MSE for linear models with normal error:
- $BIC = nlog(MSE) + klog(n)$

Likelihood Ratio Tests

LL Ratio Tests

- When two models we are comparing are nested (i.e. one is a more general version of the other), we can use likelihood ratio test to see if the increase in the LL in the general model is sufficient to justify it over the simper model.
- This is different than AIC/BIC because it is a statistical test.
- But only valid for nested models
- Test statistic: $-2log(\frac{Lik_{simple}}{Lik_{general}}) = 2 * (LL_{general} - LL_{simple})$
  Chi-squared test with degrees of freedom equal to the difference in number of parameters of general and simple model.
- Identical test stat for linear regression with normal error:
- $nlog(MSE_{simple}) - nlog(MSE_{general})$

Training MSE vs Test MSE

- $MSE = \frac{1}{n}\Sigma_i^n(y_i - \hat{f}(x_i))^2$
- We would like to minimize the "test MSE" (MSE from using our model on unseen test data) not the "training MSE" (MSE from using our model on the same data used to estimate the model)
- The next set of slides will illustrate the overfitting problem with examples and introduce the Bias Variance Tradeoff.

Training MSE vs Test MSE

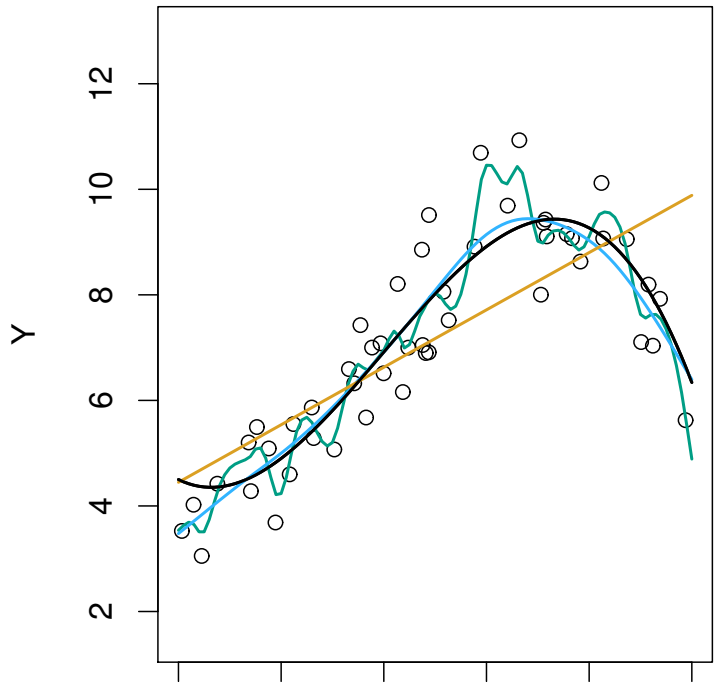
- The examples show the difference between using training MSE and test MSE to compare models (and why test MSE is preferable).
- These examples use data generated with noise. Therefore, we know the true model (unlike in the real world) and also we can generate additional data for the purpose of computing test MSE.

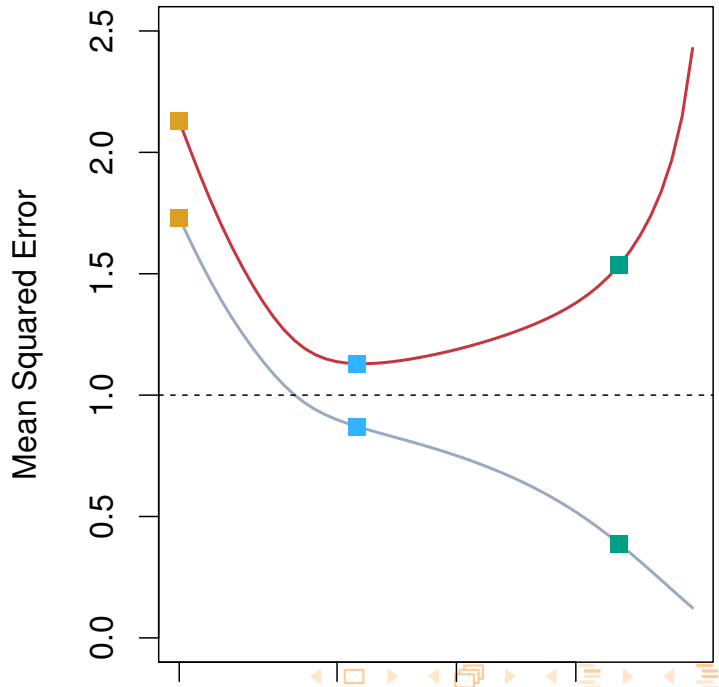
Overfitting Example (from textbook)

Black: true curve used to generate data (the correct answer!)

Dots: the training data (generated from black curve with noise)

Orange: linear fit, Blue and Green: smoothing splines





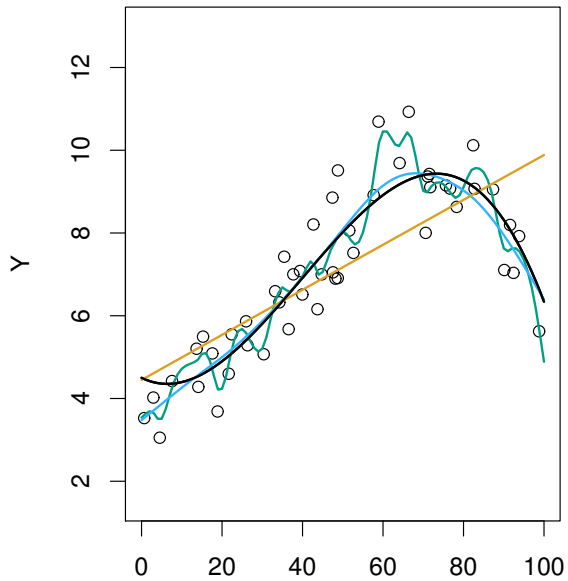
Overfitting Example (from textbook)

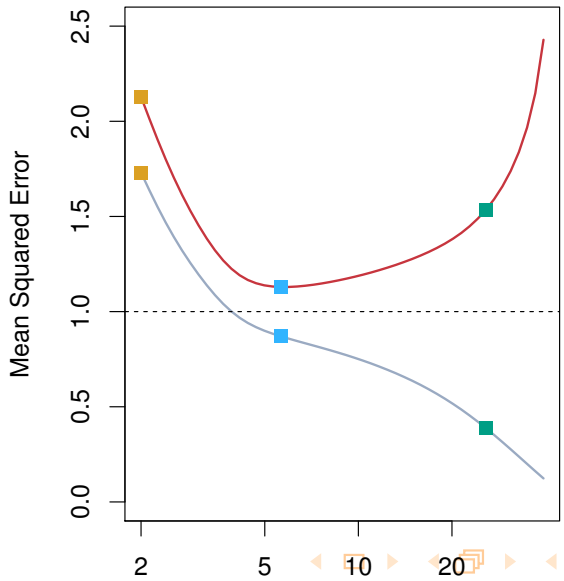
Clearly, green is best fit of data points (i.e. lowest training MSE) followed by blue then orange.

Orange is least flexible model (imposes linearity)

Green is most flexible.

More flexbility -> lower training MSE





Overfitting Example (from textbook)

Although green most closely matches the data points, blue most closely matches the true black curve.
Of our three models, blue is clearly the best (but we only know that because we know the true black curve).

Overfitting Example (from textbook)

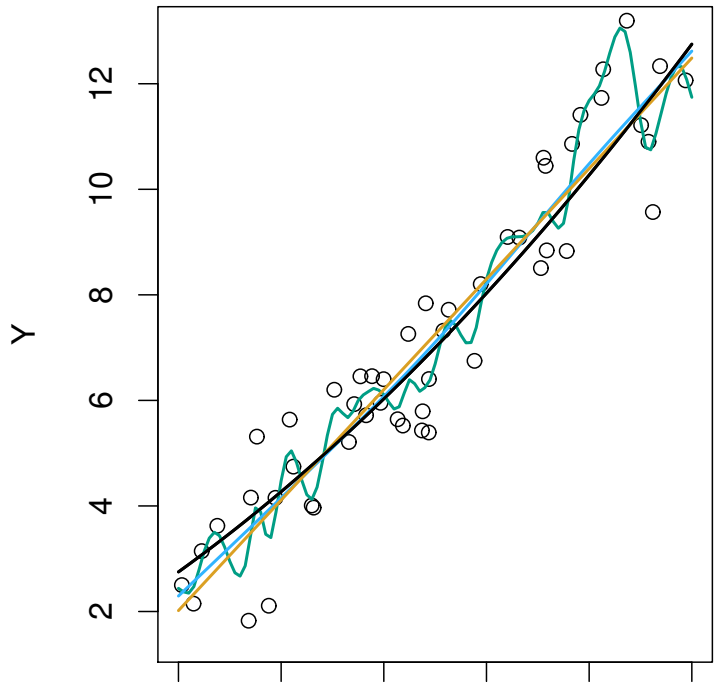
What if we compute the test MSE for our three models?
Since we can generate additional data for this example, we can create a large test data set and compute test MSE from the test data (which was not used to train).
The test MSE for the three models is shown in the right graph (in addition to training MSE).
We see that blue yields the lowest test MSE.
Typically, there is a U-shaped relationship between model flexibility and test MSE (as seen in the figure).

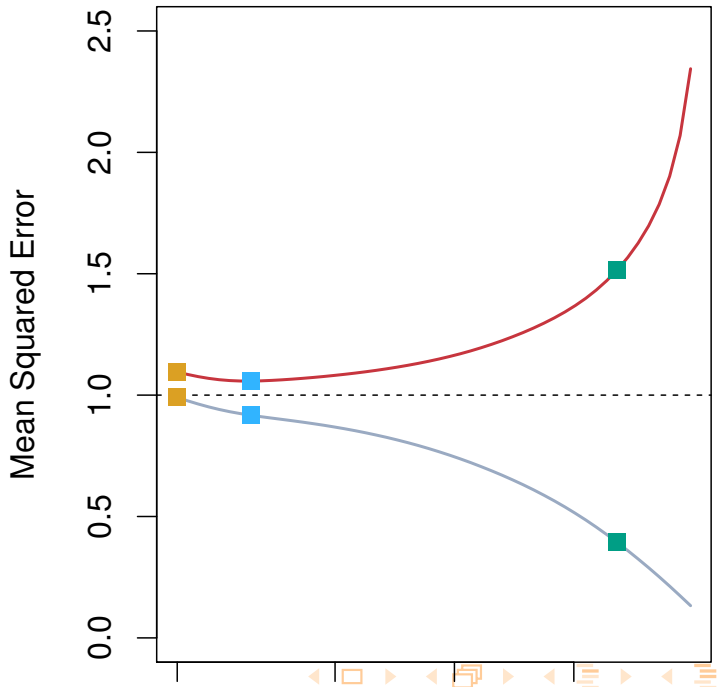
Another Overfitting Example (from textbook)

Black: true curve used to generate data

Orange: linear fit

Blue and Green: smoothing splines



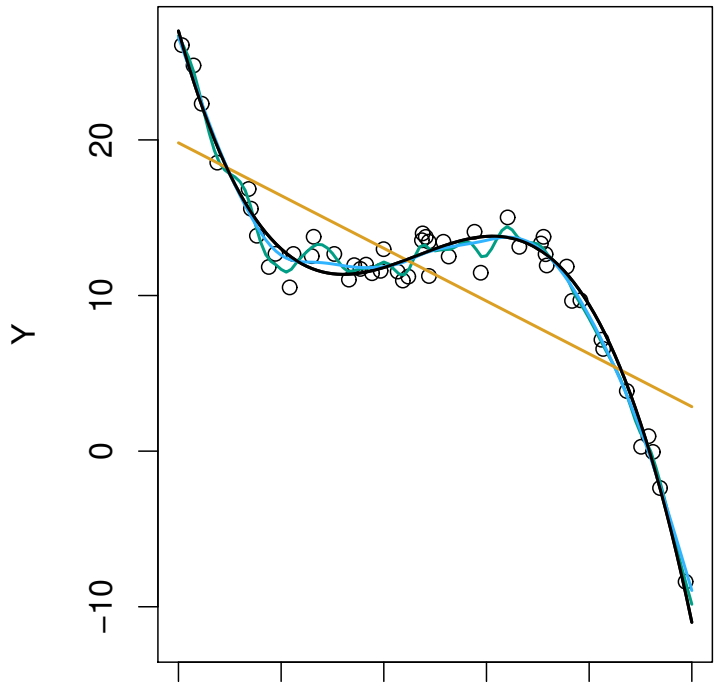


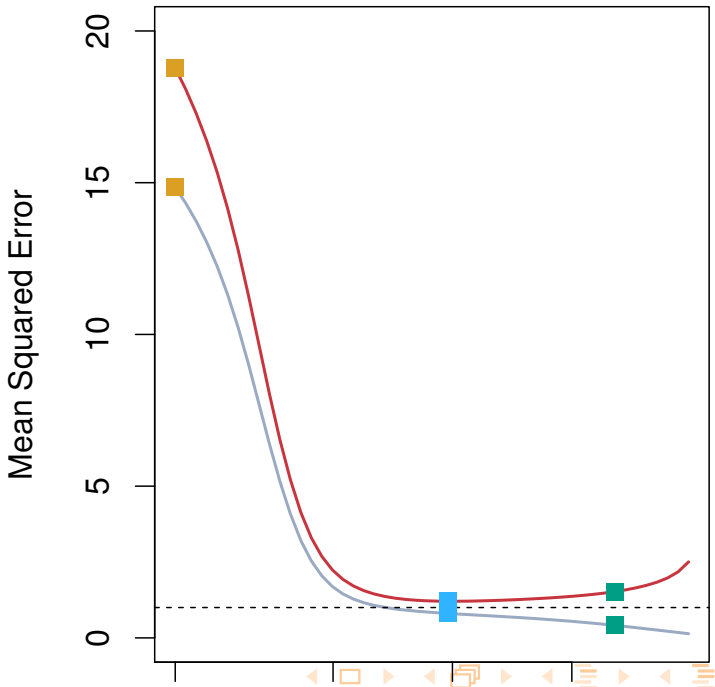
A Third Example (from textbook)

Black: true curve used to generate data

Orange: linear fit

Blue and Green: smoothing splines





Bias-Variance Tradeoff

The U shaped pattern for test MSE as model flexibility increases is explained by the bias-variance tradeoff.

$$E(y_o - \hat{f}(x_0))^2 = Var(\hat{f}(x_0)) + (Bias(\hat{f}(x_0)))^2 + Var(\epsilon) \qquad (1)$$

$Var(\epsilon)$ is the irreducible error (i.e. the error in the test observations). Even if our model is the true model, the MSE will simply be due to noise in data generation.

Bias-Variance Tradeoff

$$E(y_o - \hat{f}(x_0))^2 = Var(\hat{f}(x_0)) + (Bias(\hat{f}(x_0)))^2 + Var(\epsilon) \qquad (2)$$

Variance means the variance in model predictions if the model is estimated with different training sets.
Typically, more flexible models have greater variance.
The green curve in all examples is highly sensitive to small outliers - hence, it would change dramatically if re-estimated with a new training set. (High Variance)
The orange (linear) curve will hardly change if a few outliers are removed. (Low Variance)

Bias-Variance Tradeoff

$$E(y_o - \hat{f}(x_0))^2 = Var(\hat{f}(x_0)) + (Bias(\hat{f}(x_0)))^2 + Var(\epsilon) \qquad (3)$$

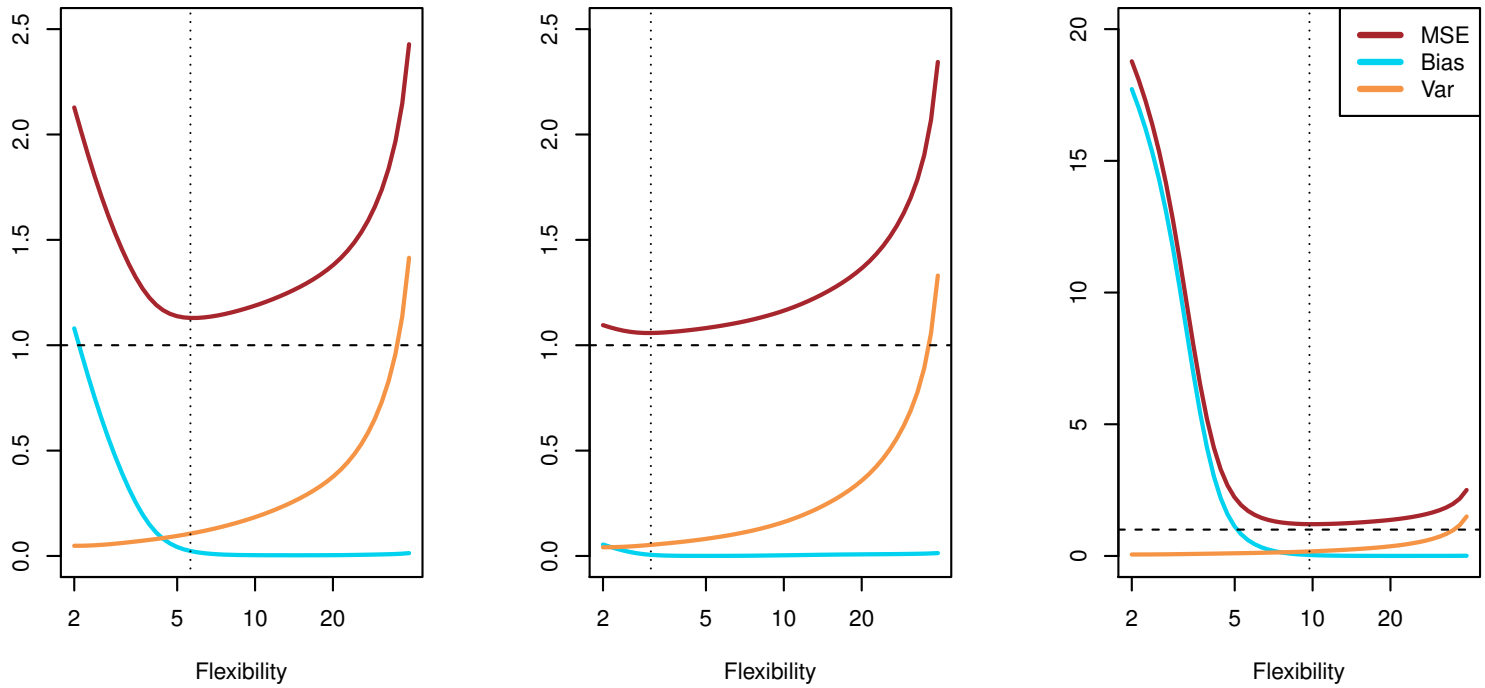
Bias refers to error that results from using an incorrect model to explain the data.
If you use a linear model to explain highly nonlinear data, no matter how large your training set, you will never get accurate predictions.
Typically, more flexible models have less bias.

Bias-Variance Tradeoff

$$E(y_o - \hat{f}(x_0))^2 = Var(\hat{f}(x_0)) + (Bias(\hat{f}(x_0)))^2 + Var(\epsilon) \qquad (4)$$

Increasing flexibility leads to less bias but more variance - this is the bias-variance tradeoff.
The bottom of the U is the best model, which balances this tradeoff.

Bias-Variance Tradeoff in the 3 Examples



Bias-Variance Tradeoff

- In practice, we do not know the true curve (i.e. the black curve) and do not have a test set.
- So there is no way to know which model minimizes the test MSE.
- These slides have shown that reducing training MSE does not always imply lower test MSE (overfitting problem).
- And, when estimating models with varying flexibility, there is a tradeoff between reducing the bias and increasing the variance.
- We will next consider cross validation, an attempt to measure test MSE.
- Then we can choose between models by picking the model with the lowest test MSE.

Cross Validation

Attempt 4: Cross Validation

- Cross Validation is an approach where the data is divided into a training set and a test set. The model is estimated with the training set and then the MSE is calculated on the test set.
- Advantage compared to previous approaches: testing predictions out-of-sample so better estimate of true test MSE
- Disadvantage: more computationally intensive because you estimate model multiple times

Validation Set

Basic Idea: split data into two parts, training set and validation set
Estimate model with training set
Use model to predict outcomes with validation set
Calculate MSE based on validation set predictions vs actual outcomes
Use this test MSE to compare models

Validation Set

Pros: really easy to implement

Cons:

1. test MSE is sensitive to choice of training set and highly variable
2. since models perform better when trained with more data, by reducing size of training set, we will over-estimate the test MSE (i.e. we could generate better out-of-sample predictions if we were training with full data set)

Leave One Out Cross Validation

Basic Idea: Estimate model with training set equal to full data set minus one observation
Use model to predict outcome for the missing observation
Calculate MSE based on prediction vs actual outcome for missing observation
Repeat above steps for every observation
Use the average MSE as test MSE to compare models

Leave One Out Cross Validation

Pros: uses almost entire data set for training so alleviates problem noted above

Cons: computationally intensive compared to validation set since esimate model n times instead of once

k-folds Cross Validation

Basic Idea: divide data set into k groups of approximately equal size
Estimate model with training set equal to full data minus the first group
Use model to predict outcomes with the omitted group
Calculate MSE based on prediction vs actual outcomes in the omitted group.
Repeate steps above for each group (k repetitions)
Use the average MSE as test MSE to compare models

k-folds Cross Validation

If $k = n$, then this is LOOCV
If $k = 2$, then this is the validation set approach.
Typically, k is set to 5 or 10 and then this approach falls somewhere between the other approaches.
1. Less computationally intensive the LOOCV but more so than the validation set approach
2. Less sensitive to the chosen training sets than valiation set but more so than LOOCV

Cross Validation with Panel Data

Suppose you have a situation where multiple individuals made choices over time (or many analogous situations involving panel data).
Any of the above cross validation methods involve splitting the data into training set and test set. Typically it makes sense to include all observations from a given individual in either the training set or the test set. i.e. do not split a given individual's data points across training and test sets.

Linear Model Selection

Reference: Chapter 6 in textbook.
We will apply the model selection techniques from earlier lecture to select the best linear model.
We will answer the question of which predictors should be included/excluded from model.
Removing irrelevant predictors from regression can improve model interpretibility.

Subset Selection

Problem: Choosing which predictors to include in regression, which to exclude.
Three possibilities:

- Best subset selection
- Forward stepwise selection
- Backward stepwise selection

Best Subset Selection

Basic Idea: Try every possible combination of predictors, select the best.

1. Let M_0 denote the null model , which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, ...p$: Fit all models that contain exactly k predictors. Pick the best among these models, and call it M_k. Here best is defined as having the smallest MSE (or largest R^2 or largest LL).
3. Select a single best model from among $M_0, ..., M_p$ using cross validation, AIC, or BIC.

Problem: As number of predictors gets large, not computationally feasible (2^n possible models).

Forward Stepwise Selection

1. Let M_0 denote the null model , which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 0, 1, 2, ..., p-1$: Consider all $p-k$ models that augment the predictors in M_k wih one additional predictor. Pick the best among these $p-k$ models, and call it M_{k+1}. Here best is defined as having the smallest MSE (or largest R^2 or largest LL).
3. Select a single best model from among $M_0, ..., M_p$ using cross validation, AIC, or BIC.

Backward Stepwise Selection

1. Let M_p denote the full model , which contains all p predictors.
2. For $k = p, p-1, ..., 1$: Consider all k models that contain all but one of the predictors in M_k, for a total of $k-1$ predictors. Pick the best among these k models, and call it M_{k-1}. Here best is defined as having the smallest MSE (or largest R^2 or largest LL).
3. Select a single best model from among $M_0, ..., M_p$ using cross validation, AIC, or BIC.